

Embedded Systems

Shelf Mind

Smart Warehouse Monitoring system

Supervisor: Dr Tamer Emara



Menna Tarek Elfar

Nada Yousry Ibrahim

Shahd Mohammed Hatab

Team members

Nourhan Mahmoud Elsherbiny

Haneen Adel Attia

Hager Mohammed Elazazy



Problem Statement

Sensitive Products

Some sensitive products such as medicines, vitamins, and skin-care products can be damaged by high temperature, humidity, sunlight, or vibration, which affects their quality and safety.

Manual Monitoring Issues

Traditional warehouses depend on manual monitoring, making it difficult to continuously check product conditions and inventory levels.

Need for a Smart System

Human errors and delayed problem detection may lead to product loss and financial damage, so an intelligent real-time monitoring system is needed.



Proposed Solution

- **Smart Monitoring System**

- Design a smart warehouse monitoring system.
- Use STM32 as the main controller.
- Automate shelf monitoring processes.

- **Inventory Management**

- Measure product quantity using load cells.
- Track shelf status in real time.
- Reduce manual inventory checking.

- **Environmental Monitoring**

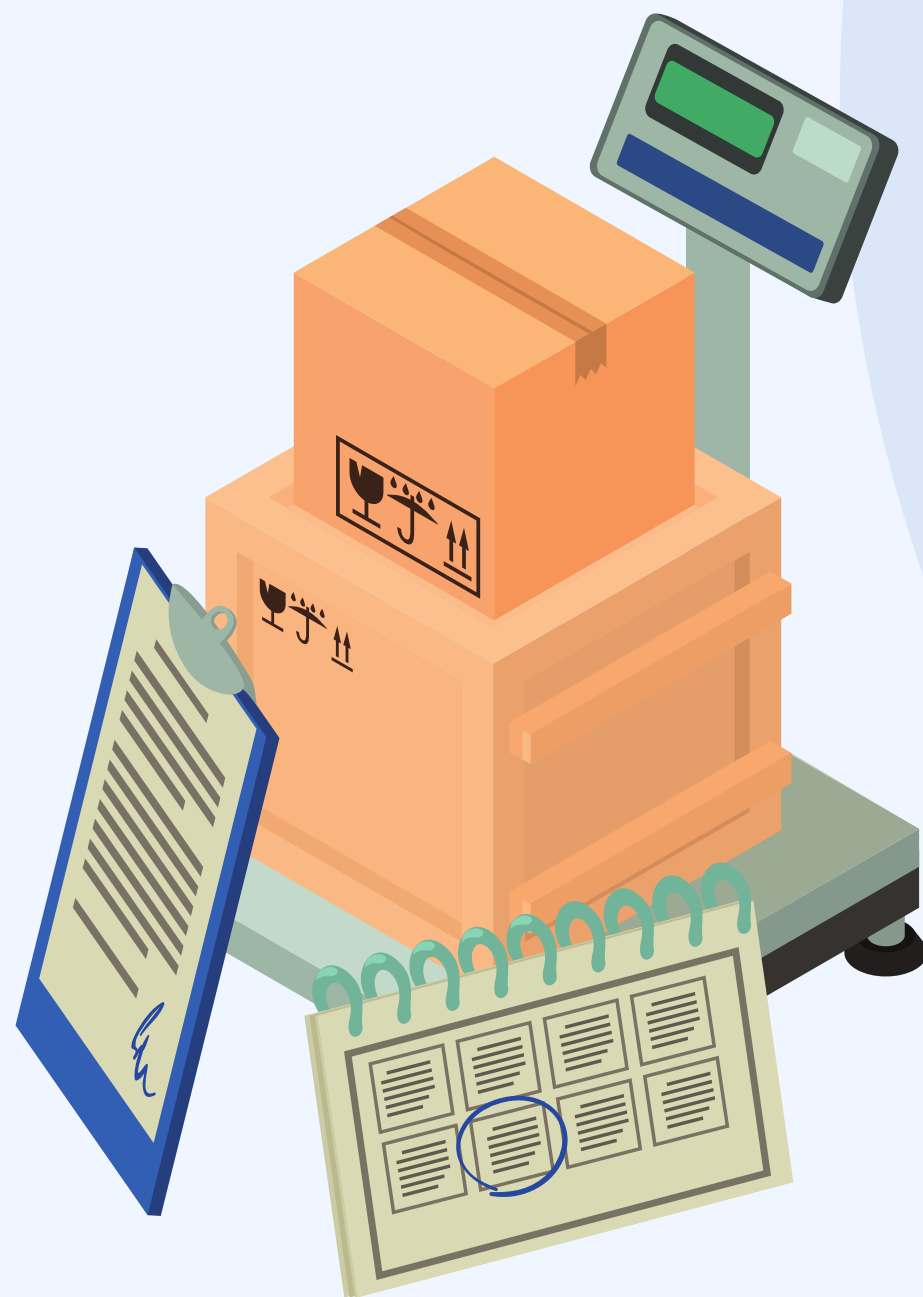
- Monitor temperature and humidity using DHT11.
- Detect sunlight exposure using LDR sensor.
- Detect vibration and movement using vibration sensor.

- **RFID Identification**

- Use RFID cards for shelf identification.
- Display product information automatically.
- Reduce human errors during monitoring.

- **Alert and Display System**

- Display live data on LCD screen.
- Use LEDs to indicate product status.
- Activate buzzer during abnormal conditions.



Our Project



Hardware Components

Main Controller

STM32 microcontroller as the central processing unit of the system.

Inventory Sensors

- HX711 with load cells to measure product weight and quantity.

Environmental Sensors

- DHT11 sensor for temperature and humidity.
- LDR sensor for light detection.
- Vibration sensor for shock and movement detection.

Identification System

- RFID RC522 module for shelf and product identification.

Output Devices

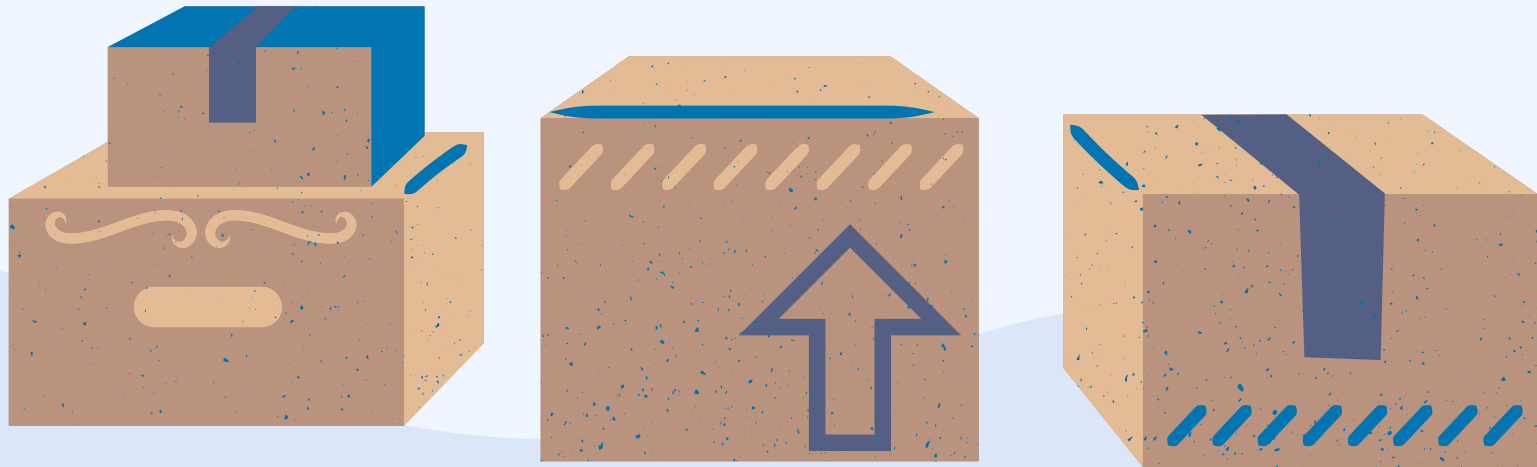
- LCD I2C display for showing real-time data.
- LEDs for status indication (green, green , red).
- Buzzer for warning alerts.

Power and Wiring

- 5V and 3.3V power supply for different modules.
- Common ground for all components.



Circuit Connections (Sensors Overview)



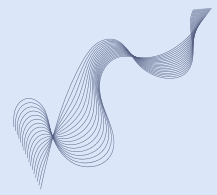
DHT11 Sensor (Temperature & Humidity)

- Type: Digital temperature and humidity sensor
- Connection: PA1 → Data pin, VCC → 5V, GND → GND
- Function: Measures environmental temperature and humidity in real time to ensure proper storage conditions, especially for sensitive products like medicines.

LDR Sensor (Light Sensor)

- Type: Digital light-dependent resistor sensor
- Connection: PA2 → Output, VCC → 5V, GND → GND
- Function: Detects sunlight or high light exposure that may affect product quality or cause damage.

Circuit Connections (Sensors Overview)



Vibration Sensor

- Type: Digital vibration/shock sensor (SW-42O or similar)
- Connection: PA8 → Output, VCC → 5V, GND → GND
- Function: Detects sudden movement or vibration of shelves, which may indicate disturbance or unsafe handling.



HX711 + Load Cells (Weight Measurement)

- Type: Load cell amplifier + strain gauge weight sensor
- Connection: Shelf 1: DT PA11, SCK PA12
- Shelf 2: DT PB12, SCK PB13
- Shelf 3: DT PA15, SCK PB3
- VCC → 5V, GND → GND
- Function: Measures product weight to monitor stock level and detect missing or reduced quantities in real time.



Circuit Connections (Sensors Overview)



RC522 RFID Module

- Type: RFID reader (13.56 MHz communication module)
- Connection:
 - SDA → PA4
 - SCK → PA5
 - MOSI → PA7
 - MISO → PA6
 - RST → PA3
 - VCC → 3.3V
 - GND → GND
- Function: Identifies shelves/products using RFID cards to display correct product information and track each shelf separately.

Circuit Connections (Sensors Overview)

Buzzer

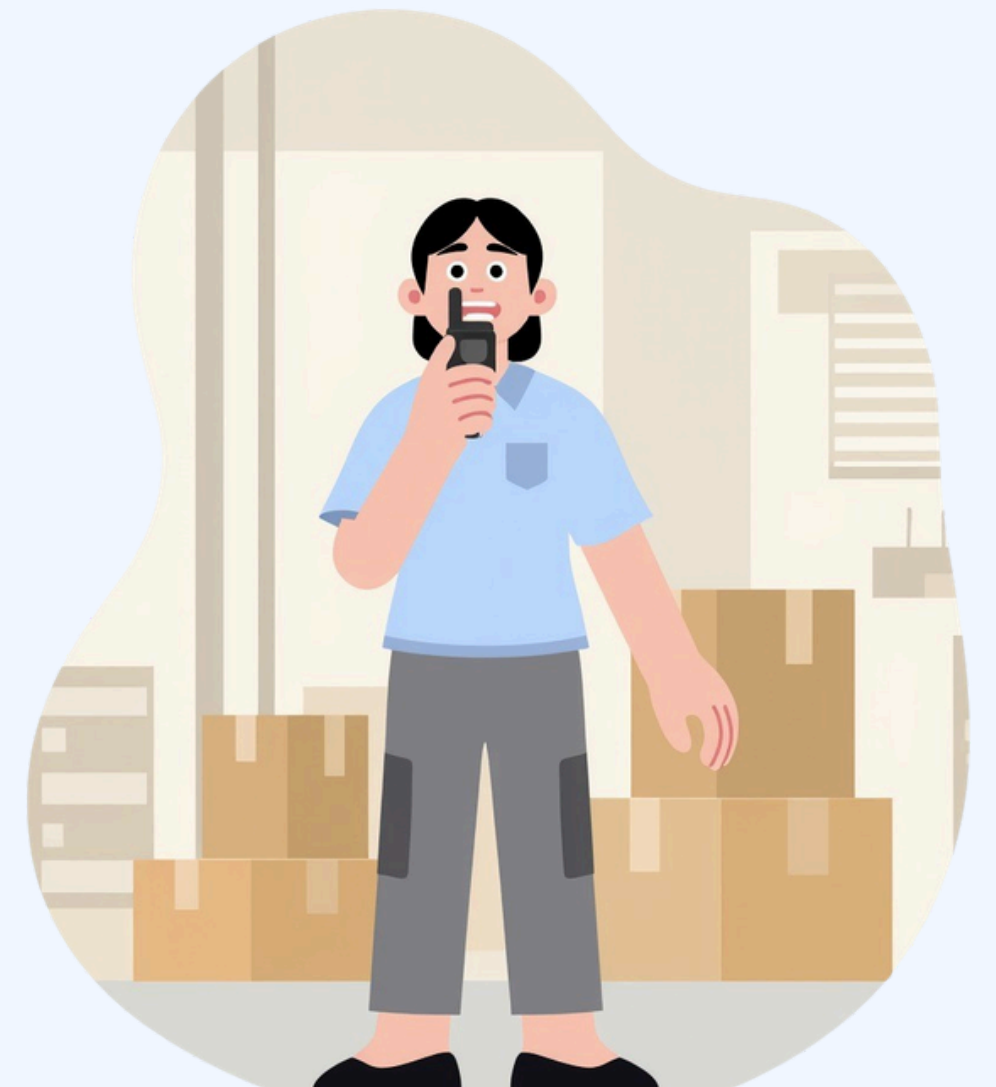
Type: Active buzzer (audio alert device)

Connection: PAO → Control pin, VCC → 5V, GND → GND

Function: Generates an audible alarm when abnormal conditions are detected such as high temperature, high humidity, excessive vibration, or stock shortage, providing immediate real-time warning.

LEDs

- Type: Status indicator LEDs (Green / Green / Red)
- Connection:
- Green LED → PBO
- Green LED → PB11
- Red LED → PB1
- VCC → 5V
- GND → GND
- Function: Indicate system status visually: Green for normal conditions, Green for warning states, and Red for critical alerts such as unsafe environmental conditions or inventory issues, enabling quick visual monitoring of the warehouse status.



Circuit Connections (Sensors Overview)



LCD Display (I2C)

- Type: 16x2 I2C LCD Display
- Connection:
 - SDA → PB7
 - SCL → PB6
 - VCC → 5V
 - GND → GND
- Function: Displays real-time system data including temperature, humidity, shelf weight, product name, and alert messages to provide clear and easy monitoring of the warehouse status.



Code Core

```
void setup() {  
  
  Wire.begin(PB7, PB6);  
  
  lcd.init();  
  lcd.backlight();  
  
  dht.begin();  
  
  SPI.begin();  
  mfrc522.PCD_Init();  
  
  scale1.begin(DT1, SCK1);  
  scale2.begin(DT2, SCK2);  
  scale3.begin(DT3, SCK3);  
  
  scale1.set_scale(cal1); scale1.tare();  
  scale2.set_scale(cal2); scale2.tare();  
  scale3.set_scale(cal3); scale3.tare();  
}
```

System Initialization

- Explanation
- Initializes all sensors and communication protocols
- Starts LCD, RFID, DHT11, and HX711 modules
- Calibrates all load cells before operation



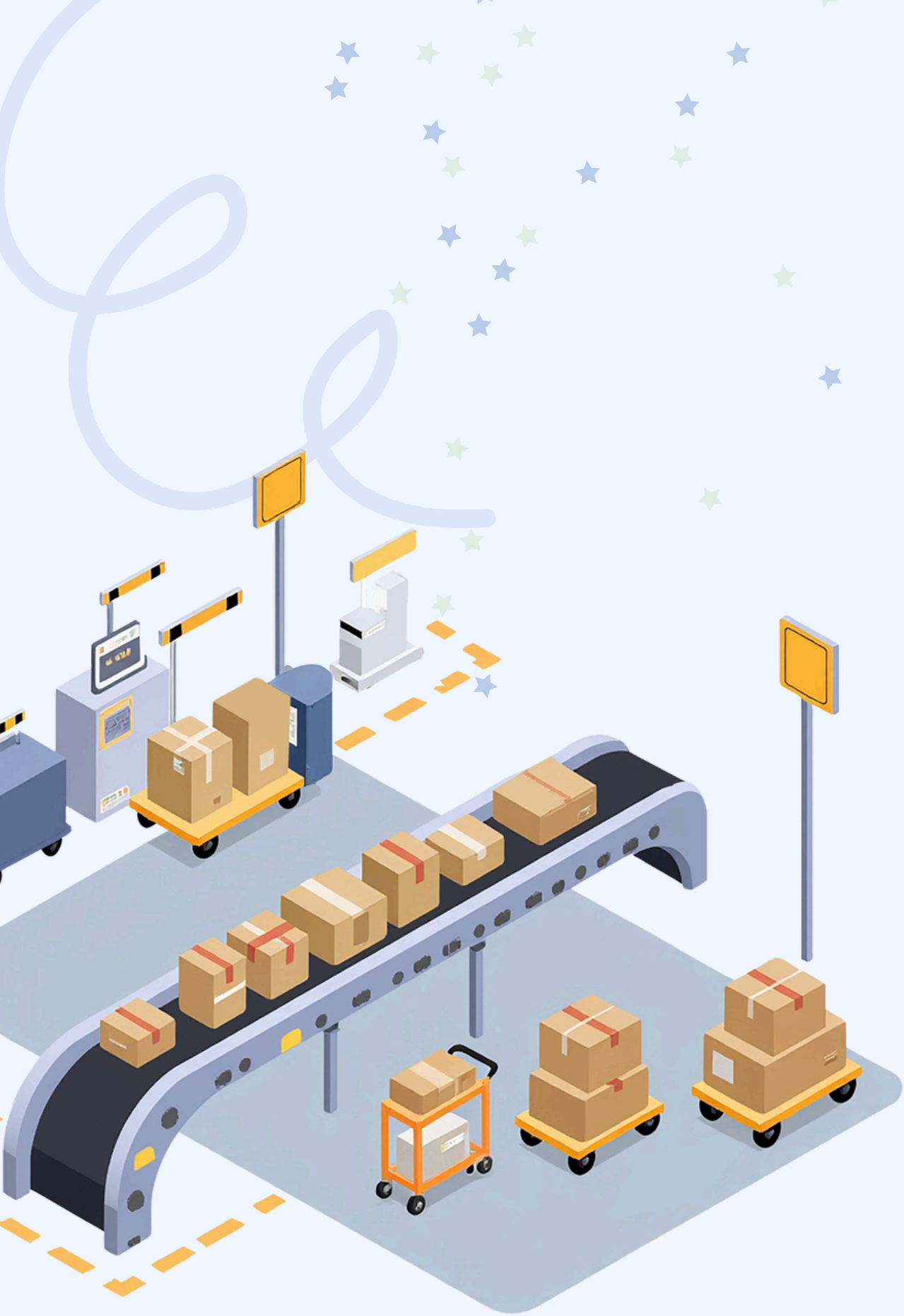
Code Core

RFID Identification

```
if (mfrc522.PICC_IsNewCardPresent() && mfrc522.PICC_ReadCardSerial()) {  
  
    String uid = getUID(mfrc522.uid.uidByte, mfrc522.uid.size);  
  
    int shelf = -1;  
  
    if(uid == "D4555A06") shelf = 0;  
    if(uid == "9354AAD9") shelf = 1;  
    if(uid == "1A216F06") shelf = 2;  
}
```

Explanation

- Detect RFID card
- Read UID
- Assign each UID to a shelf



Sensor Data Reading

```
if (millis() - lastReadTime > 2000) {  
  
    lastReadTime = millis();  
  
    humidity = dht.readHumidity();  
    temperature = dht.readTemperature();  
}  
  
bool light = digitalRead(LDR_PIN);  
bool vib = digitalRead(VIB_PIN);
```

```
float weight = 0;  
  
if(shelf==0) weight = readWeight(scale1);  
if(shelf==1) weight = readWeight(scale2);  
if(shelf==2) weight = readWeight(scale3);
```

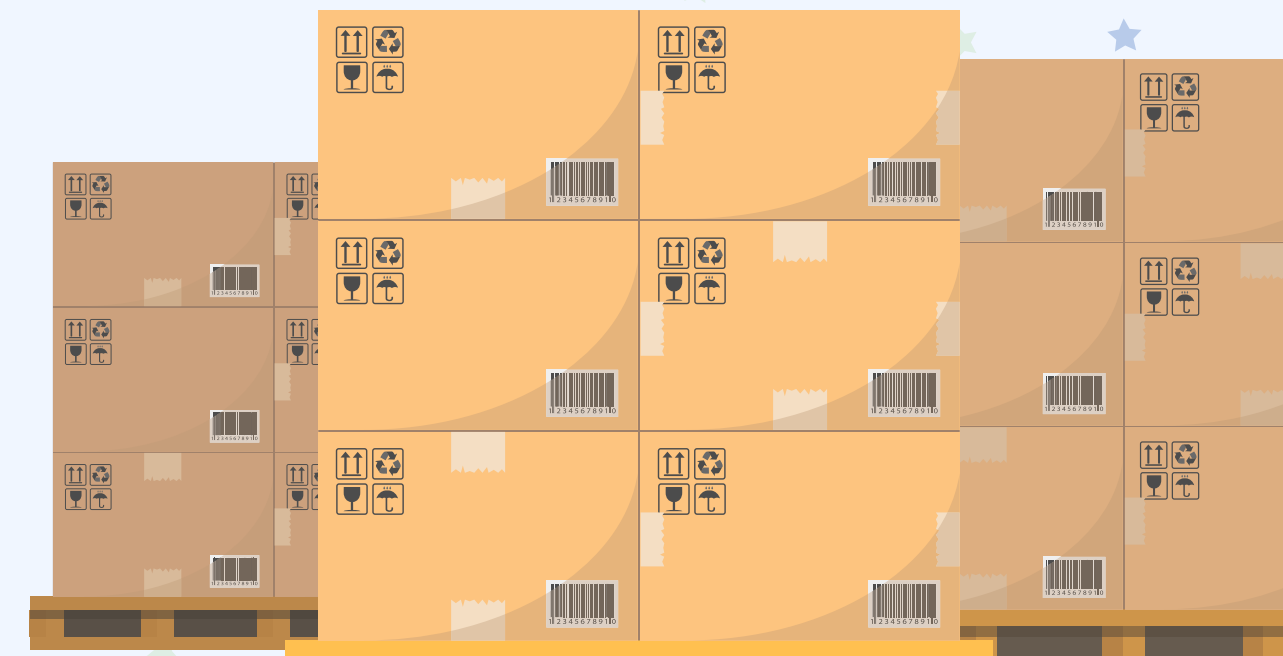
Code Core

Explanation

- DHT11 → Temperature & Humidity monitoring
- LDR → Light / sunlight detection
- Vibration sensor → Detects shocks or movement

Explanation

- Each shelf has its own load cell
- HX711 converts analog weight signals to digital
- Used for inventory quantity tracking



Code Core

Real-Time Display

Explanation

- Shows shelf number
- Displays product name
- Shows weight, temperature, humidity
- Provides real-time monitoring

Warning Logic

Explanation

- Green / Yellow / Red LED indicators
- Buzzer activates on abnormal conditions
- Immediate warning system

```
lcd.clear();  
lcd.setCursor(0,0);  
lcd.print("SHELF ");  
lcd.print(shelf+1);  
  
lcd.print(products[shelf]);  
lcd.print(weight);  
lcd.print("g");  
  
lcd.print("T:");  
lcd.print(t2);  
lcd.print("H:");  
lcd.print(h2);
```

```
if(ok){  
    lcd.print("PRODUCT OK");  
}  
else{  
    lcd.print("PRODUCT ERROR");  
    beep(150);  
}
```

Conclusion



Future Work

Possible Improvements

- Add IoT cloud system for remote monitoring.
- Develop mobile application dashboard.
- Store data in database for analytics.
- Increase number of shelves and scalability.
- Add AI-based prediction for product spoilage.

Conclusion

- The system provides a smart and automated warehouse solution.
- It ensures real-time monitoring of environmental conditions and inventory.
- It improves product safety, reduces losses, and increases efficiency.
- It demonstrates how IoT and embedded systems can modernize warehouses.

**Thank you for
your attention**

